

Empirical design for emergency unemployment assistance

Australians and New Zealanders in shared labour markets: promise, threats and a journal-ready redesign

Labour economics teaching deck | Based on manuscript v10 and structural model 15 | August 2026



What this deck is designed to teach

Understand the estimand

Separate eligibility for the March policy bundle, realised receipt and the \$550 supplement as distinct causal objects.

Audit the design

Trace selection, risk-set, timing, policy-interaction, measurement and one-shock inference threats.

Build the next version

Specify a pre-policy cohort, person-week hazard design, conservative uncertainty package and decisive exhibit set.

The institutional comparison is genuinely promising. The empirical rebuild—not extra prose—is the gate to a strong public-economics submission.

Bottom line on the current empirical design

Why it is attractive

Linked weekly payroll, tax, visa and benefit records; large benefit change; ineligible New Zealand citizens living in the same Australian labour markets.

Why it is not ready

Treatment uses realised future benefit receipt; samples condition on future non-employment; inference treats 48 cells as IID; JobKeeper and the policy bundle remain confounded.

What would change the verdict

Stable estimates after a pre-policy cohort rebuild, correct risk sets, long pre-trends, design-specific inference and direct policy-interaction checks.

Current point estimates are useful targets and descriptive facts. They are not yet journal-ready causal estimates.

Roadmap

Part I — Learn the current design

Question and estimand
Institutional comparison
Data and matching
Weekly outcomes
DiD and current estimates

Part II — Rebuild for submission

Post-treatment selection
Risk sets and person-week estimation
One-policy inference
Policy bundle and JobKeeper
Heterogeneity, earnings and paper exhibits

The deck treats shortcomings as design decisions with concrete repairs, not as a generic robustness checklist.

01

Question, estimand and institutional experiment

A durable public-finance question comes before an estimator.

The durable question is broader than one pandemic programme

Policy question

How does a large increase in unemployment assistance change movements into and out of employment when jobs are scarce, health risks are salient and search requirements are relaxed?

Why labour economists care

- Search and acceptance responses
- Liquidity versus moral hazard
- Worker versus firm-driven exits
- State dependence of UI effects
- Interactions with wage subsidies and conditionality

Frame Australia–New Zealand as the identification design; frame emergency benefit design as the contribution.

Three plausible estimands are currently mixed together

Estimand	Treatment definition	Interpretation	Status
Eligibility / policy-bundle ITT	Nationality × pre-policy legal eligibility	Effect of exposure to the March JobSeeker expansion	Preferred main estimand
Realised receipt effect	Received JobSeeker at any time in 2020	Effect among endogenous takers / job losers	Post-treatment; secondary only
\$550 supplement alone	Incremental cash amount holding other reforms fixed	Pure benefit-level effect	Not isolated by baseline design
Instrumented receipt	Eligibility instrument for actual receipt	Local effect under exclusion/monotonicity	Possible only with credible first stage and exclusion

Recommended sentence: estimate exposure to Australia's March 2020 JobSeeker expansion relative to ineligible New Zealand citizens in shared local labour markets.

The comparison group is unusual—and economically informative

Australian citizens

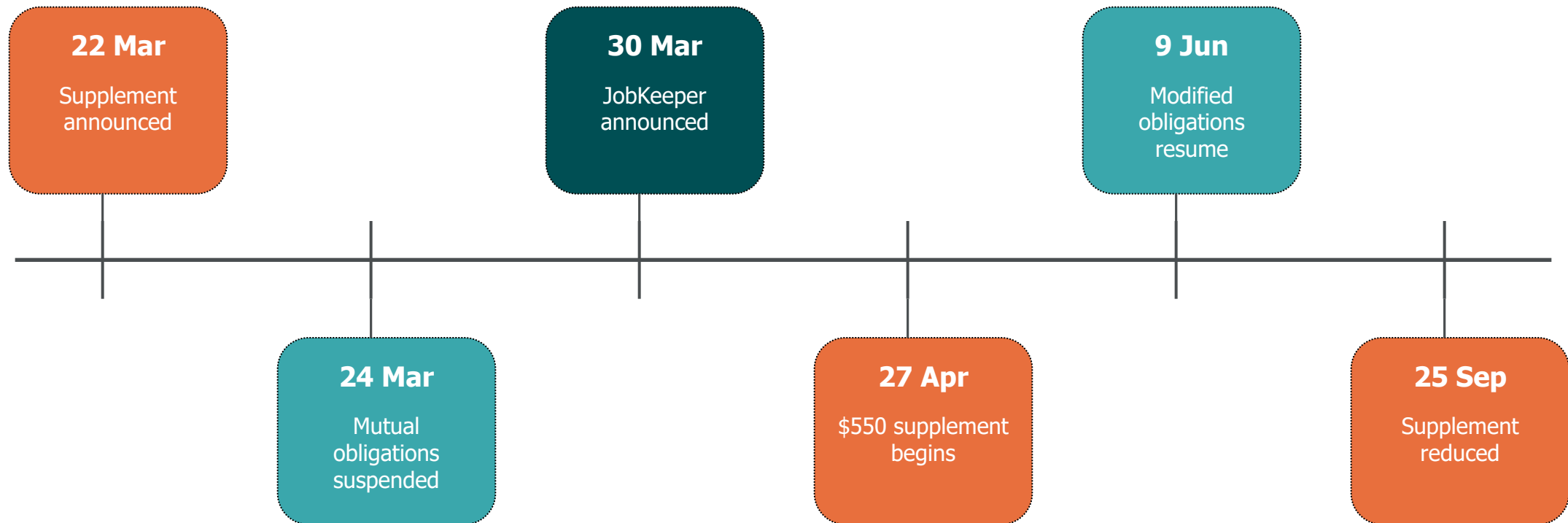
Potentially eligible for JobSeeker and the supplement
Also affected by eligibility expansions, waivers and mutual-obligation changes
Actual receipt depends on job loss and take-up

New Zealand citizens on SCV

Live and work in Australia
Generally ineligible for JobSeeker within the selected residency window
Face many common local shocks but may differ in networks, tenure, mobility and selection

Say «shared local shocks conditional on observables and parallel trends»—not «identical conditions».

The «treatment date» is a sequence of announcements and implementation dates



Estimate announcement, entitlement and payment windows separately where feasible; otherwise label the estimand as exposure to the March JobSeeker policy bundle.

The reduced-form treatment is a policy bundle

Component	Timing / exposure	Why it matters for outcomes	Can baseline isolate it?
Coronavirus Supplement	Announced 22 Mar; paid from 27 Apr	\$550/fortnight changes outside option	No—coincides with other changes
Eligibility and means-test changes	Implemented in March	Changes who can receive JobSeeker	No—part of nationality-linked exposure
Waiting-period / asset-test waivers	March 2020	Changes access speed and effective generosity	No
Mutual obligations	Suspended then staged return	Changes search cost/requirements	Potential triple-difference variation
JobKeeper wage subsidy	Announced 30 Mar	Directly affects retention and vacancies	Needs actual/predicted exposure
Health restrictions	National + state timing	Affect demand, risk and feasible work	Common only imperfectly

Main causal label should be «JobSeeker expansion / March policy bundle» unless component-specific variation is demonstrated.

02

What the current design does

Linked data, realised-receipt matching, selected weekly risk sets and a two-group DiD.

Administrative data are a major comparative advantage

Source	Role in design	Key validation question
Single Touch Payroll	Weekly pay/employment histories and transitions	Does no pay mean no employment? How are late/irregular reports handled?
Visa / Home Affairs	Identify New Zealand citizens and residency tenure	Who is legally ineligible and continuously resident?
DSS / DOMINO	Identify JobSeeker receipt	Receipt is post-treatment; can pre-policy statutory eligibility be reconstructed?
Personal income tax	Income, occupation, partnership and annual outcomes	Can pre-period wages and actual replacement rates be built?
BLADE / business tax	Industry and employer characteristics	Can JobKeeper eligibility/receipt and firm exit be linked?

The data can support a much stronger design if the raw-to-person-week construction layer is rebuilt and documented.

Current sample construction uses future outcomes twice

Current treatment/sample

Australian treatment: received JobSeeker at any time in 2020.

Both groups: at least four consecutive weeks out of work in 2020.

Matched sample $\approx 22,873$ AU + $22,859$ NZ.

Current exit risk

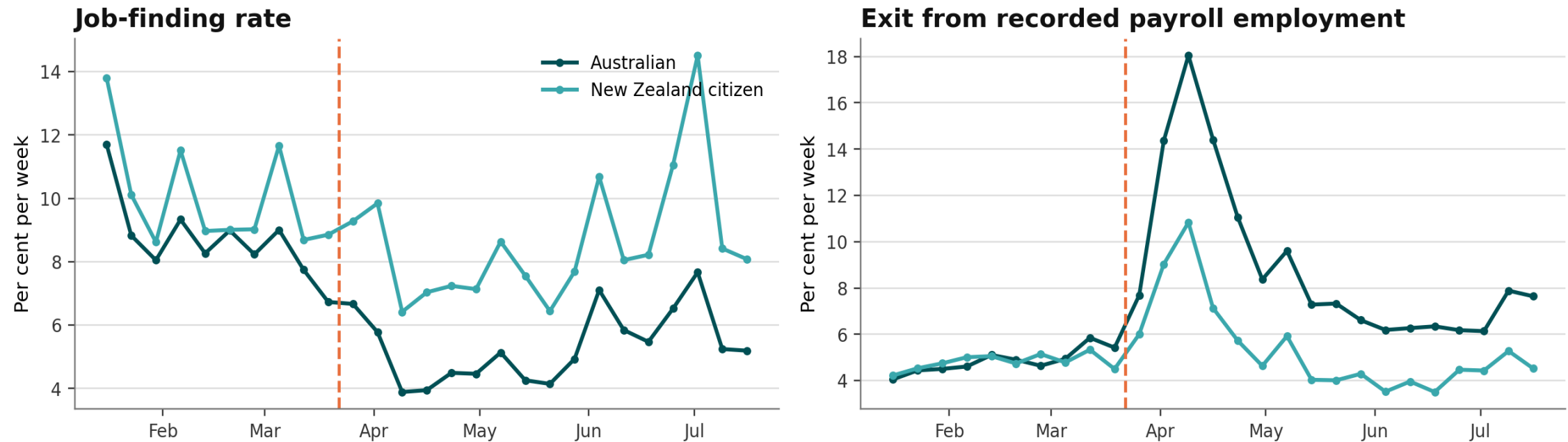
Employment-exit analysis is drawn from people known to experience non-employment later in 2020.

A policy-induced exit can add someone to the sample; a prevented exit can remove them.

This changes both composition and the meaning of the denominators. It is not repaired by more controls or propensity-score matching.

Current weekly patterns are striking—but inherit selected denominators

Current cleared weekly series: informative patterns, selected risk sets

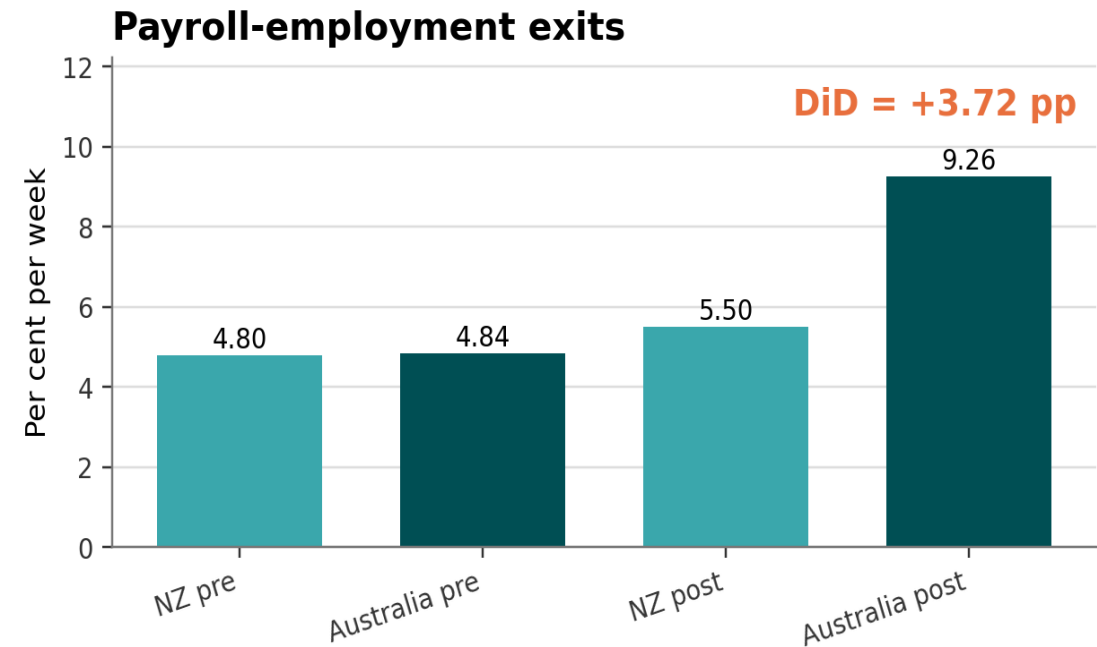
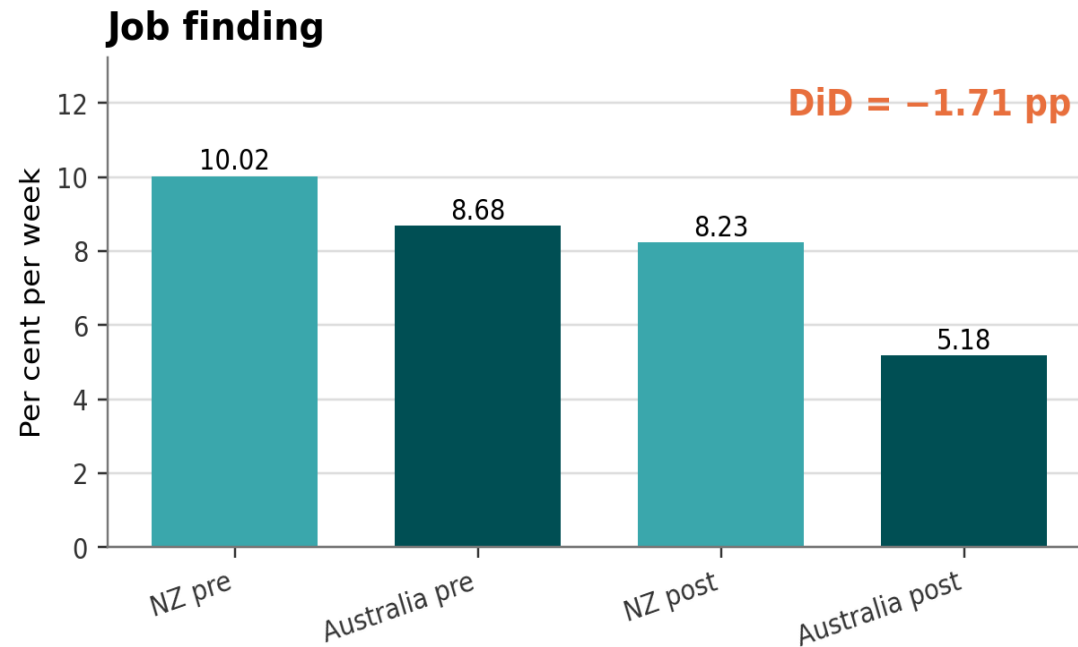


Dashed line: 22 March announcement. These series inherit the draft's realised-receipt and future non-employment selection rules.

Use these as descriptive motivation while rebuilding the cohort and risk sets.

Current pre/post cells imply large differential changes

The draft's four empirical cells discipline model 15



Source: manuscript Tables 3–4 / model-15 target vector. Percentage-point effects use Australian minus New Zealand changes.

Primary reporting should use percentage points, named baseline rates and uncertainty—not only «19%» or «64%».

The baseline two-group DiD is transparent

Equation

$$Y_{gt} = \alpha + \beta_1 \text{Australian}_g + \beta_2 \text{Post}_t + \beta_3 (\text{Australian}_g \times \text{Post}_t) + \varepsilon_{gt}$$

$$\beta_3 = [\text{AU post-pre}] - [\text{NZ post-pre}].$$

Causal requirement

Absent the policy exposure, the adjusted Australian weekly transition path would have changed like the comparable NZ path.

No other nationality-differential shock occurs at the same time.

The equation is simple; the hard work is cohort definition, policy isolation and uncertainty for one treated contrast.

Current headline estimates—and why precision is provisional

Outcome	Draft DiD	Baseline	Draft uncertainty	Immediate caution
Job finding	−1.71 pp	AU pre 8.68% weekly	SE 0.82 pp; $p=0.043$	48 nationality-week cells; IID errors
Payroll employment exits	+3.72 pp	AU pre 4.84% weekly	SE 1.40 pp; $p=0.011$	Selected exit sample; IID errors
Earnings / match quality	Claimed no gain	Annual descriptive indices	No causal interval in reviewed script	Rebuild formal outcomes or remove claim
JobKeeper-adjusted exits	+2.60 pp	Same selected series	Single month dummy	Not a measure of actual exposure

Do not lock the abstract's 19% and 64% claims until the redesigned denominators and inference are final.

Why the design is still worth rebuilding

Measurement

Weekly administrative transitions can measure flows and spell dynamics much more sharply than household surveys.

Counterfactual

NZ citizens share Australian local labour markets yet differ in benefit access—a rare comparison for an aggregate emergency reform.

Policy relevance

The flat, large supplement generates meaningful variation in predicted replacement-rate changes and household exposure.

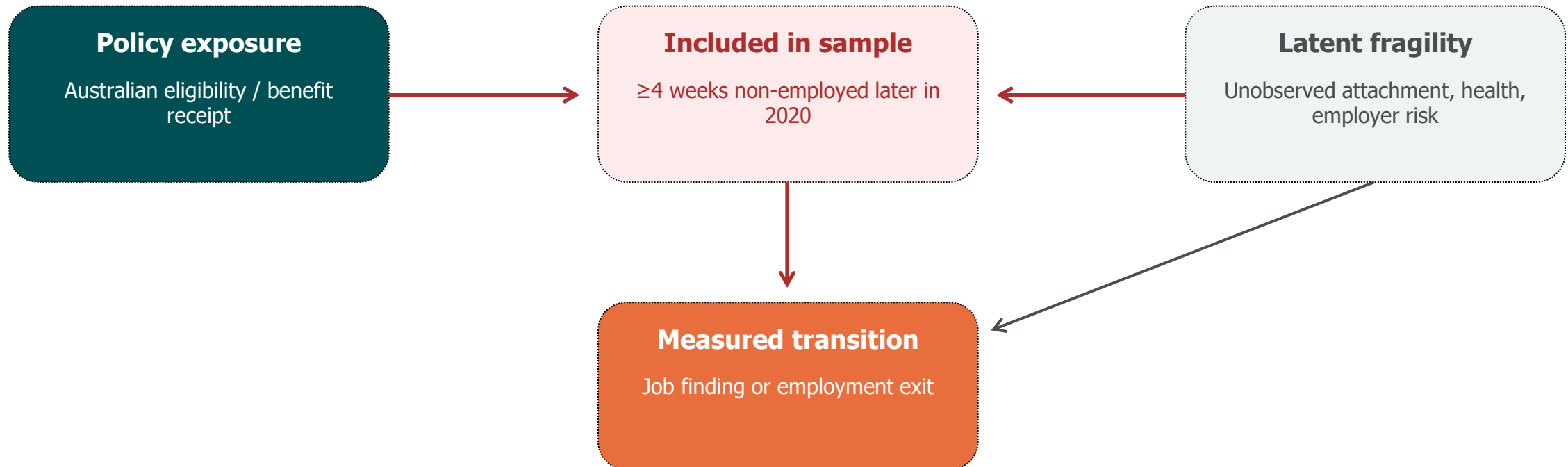
If the revised results survive, the design can speak to state-contingent unemployment assistance beyond COVID-19.

03

Threats to identification and inference

The central issues are selection, one-shock uncertainty and bundled interventions.

Conditioning on future non-employment creates a collider



Selecting on the middle box opens a non-causal path between policy exposure and latent fragility. The exit sample is especially mechanical because inclusion requires a later exit.

Realised receipt is an outcome of the same process being studied

Why it is endogenous

Receipt depends on job loss, take-up, claim timing, partner income, assets and administration.

Job finding and employment exits directly affect the chance of becoming a recipient.

Preferred treatment

Citizenship $\times$ pre-policy legal eligibility.

If feasible, predict 1 March eligibility from only pre-period information.

Use actual receipt as a mediator or instrumented secondary analysis.

Eligibility-based ITT sacrifices a «treated recipients» label but gains a coherent causal estimand.

One policy shock does not create 48 independent experiments

Current problem

Weekly outcomes are serially correlated.
Treatment turns on once for one nationality group.
IID standard errors treat cell residuals as independent.
Clustering on people cannot create more policy shocks.

Implication

The effective design uncertainty is about the Australian-minus-NZ time series around one event.
Confidence intervals may be materially wider than conventional cell-level or person-clustered errors.

Administrative N improves outcome measurement and adjustment; it does not solve few-policy inference.

A transparent inference package for one treated contrast

Method	What it preserves	Role / caveat
Pre/post collapse	Low-frequency policy contrast	Conservative benchmark; few observations
HAC on weekly AU–NZ gap	Serial correlation over calendar time	Show bandwidth and small-sample sensitivity
Block bootstrap	Within-series dependence	Resample calendar/residual blocks, not individual weeks
Pair/design bootstrap	Matching and person composition	Rebuild weights/risk sets within each draw
Placebo-date distribution	How unusual the break is in admissible pre-periods	Reference distribution; dates are not literally random
Multiple placebo years	Seasonality and recurring nationality gaps	Must use identical cohort/risk-set algorithm

Report intervals from several defensible methods. Do not use significance stars as the core argument.

A short flat pre-period has low power against damaging trend differences

- Extend consistently into 2019 as far as payroll coverage permits.
- Plot raw and composition-adjusted Australian-minus-NZ gaps.
- Estimate event-study coefficients and joint pre-period tests, but do not treat non-rejection as proof.
- Report the minimum detectable pre-trend and how much differential trend overturns the conclusion.
- Apply Rambachan–Roth / HonestDiD sensitivity to bounded deviations from parallel trends.
- Use nationality-specific trends only as a sensitivity check; short-window extrapolation is model-dependent.

The identifying assumption is a counterfactual path claim. Balance and a pre-trend p-value are evidence—not validation.

JobKeeper is a direct retention treatment, not an April nuisance dummy

Why it confounds

JobKeeper preserves worker-firm links, including zero-hours jobs.

It changes measured exits, vacancy creation and competition for new jobs.

Its interaction with JobSeeker may differ by nationality.

Preferred strategies

Link actual receipt / firm eligibility.

Split supported and unsupported jobs.

Use pre-pandemic predicted exposure or firm eligibility designs.

Restrict to clearly ineligible firms as a bounded sample.

If exposure cannot be measured, label residual confounding honestly—especially for employment exits.

Payroll records are powerful, but missing pay is not always non-employment

Potential false transitions

- Fortnightly or irregular payment cadence
- Late reports and corrections
- Zero-hours retention
- Employer STP entry/exit
- Concurrent jobs
- Self-employment not recorded

Validation design

- 1-, 2- and 4-week persistence
- Fortnightly aggregation
- Regular-payer restriction
- Quarterly/annual tax cross-checks
- Any-job versus main-job states
- Employer and person exit flags

Call the outcome «exit from recorded payroll employment» unless voluntary quits and layoffs can be separated.

Nationality comparability remains the substantive counterfactual assumption

Plausible shared shocks

- Same national macroeconomy
- Same state/local restrictions
- Often same industries and occupations
- Common payroll measurement system

Possible differential responses

- Migration tenure and networks
- Household composition
- Return-migration options
- Industry/employer sorting
- Risk tolerance and work orientation
- Access to other support

Match/reweight on observables and pre-outcomes; test heterogeneity and mobility; state conditional parallel trends explicitly.

Matching needs diagnostics, not only «balanced distributions»

Required diagnostic	What it reveals	Decision rule
Standardised mean differences	Balance for every covariate and key interaction	Pre-specify acceptable threshold
Variance ratios	Distributional imbalance beyond means	Report before/after
Propensity overlap + trimmed share	Where counterfactual support exists	Restrict to common support
Pre-policy outcome balance	Whether transition histories align	Include employment, JFR, exits, duration, earnings
Caliper / replacement sensitivity	Dependence on matching algorithm	Compare with overlap/entropy weights
Matched-pair identifiers	Dependence and design structure	Retain in estimation and bootstrap

Do not select the matching method by the most significant treatment effect. Matching improves comparability; it does not randomise nationality.

04

A journal-ready re-estimation plan

Pre-policy cohorts, contemporaneous hazards, person-week models and pre-specified validation.

The redesign starts with a pre-policy cohort and weekly risk sets



Job-finding risk set

Everyone not in recorded payroll employment in week t .

Add flexible duration dependence.

Employment-exit risk set

Everyone in recorded payroll employment in week t .

Persistence is an outcome definition—not inclusion.

Run 1-, 2-, and 4-week persistence definitions; report weekly denominators, events and censoring by nationality.

Preferred person-week specifications

Linear probability baseline

$Y_{i,t+1} = \text{week FE} + \beta(\text{AU}_i \times \text{Post}_t)$
+ pre-X weights/interactions + duration FE + ε_{it}

Easy to interpret in percentage points; estimate on weekly risk sets.

Discrete-time hazard complement

Logit or complementary log-log link
Flexible duration dependence
Calendar-week effects
Matched-pair or weighting structure
Average marginal effects for comparability

Cell-level AU–NZ series remains essential for transparency and time-series inference; person-week estimation adds covariate and duration control.

Covariates should respect timing and risk-set mechanics

- Calendar-week fixed effects.
- Only pre-treatment demographic, household, industry, occupation, earnings and attachment measures.
- Non-employment-duration fixed effects for job finding.
- Job tenure or pre-policy attachment controls for employment exit where available.
- Matched-pair fixed effects or design weights if matching remains central.
- Region-by-week and industry-by-week controls only after checking they do not absorb the sole useful contrast.
- Payroll cadence and employer coverage indicators.
- Exact person, employer, person-week and event counts in every table.

The preferred specification should be chosen for design coherence—not because it maximises β /SE.

Event-study design should diagnose dynamics, not certify parallel trends

Main figure

- Weekly AU–NZ adjusted gap
- One omitted pre-period
- Long 2019 pre-window
- Announcement and payment markers
- Confidence bands using design-specific uncertainty

Sensitivity overlays

- Alternative cohort/risk-set definitions
- HonestDiD bounds
- JobKeeper exposure split
- Leave-one-industry/region/age-out
- Fortnightly aggregation
- Post-taper and expiry dynamics

A good event study makes the identifying comparison visually auditable week by week.

Mutual obligations require a direct pooled triple difference

Current approach

Run separate Victoria and non-Victoria DiDs around July.
One insignificant coefficient and one significant coefficient do not imply the coefficients differ.

Required test

Estimate $AU \times Post \times NonVictoria$ with all lower-order terms.
Test the triple interaction directly.
Plot dynamics and confront Victoria's distinct outbreak/restrictions.

Interpret modestly: differential nationality gaps when requirements resumed—not a clean proof of «motivation» or zero conditionality effects.

Heterogeneity should use actual predicted replacement-rate changes

Current problem

Income-bin DiDs are -2.37 , -4.25 and -6.04 pp from low to high income.

The model file relabels these as low/mid/high replacement rates using placeholder rates.

That mapping is not valid.

Required construction

Predict statutory pre/post entitlement from pre-policy household information.

Divide by a clearly defined pre-displacement wage.

Estimate continuous/bin/spline interactions and direct equality/trend tests.

Show absolute benefit changes and proportional replacement-rate changes; separate them from sector, age and household exposure.

The earnings / match-quality claim needs a real treatment-effect design

To retain the claim

Unconditional earnings including zeros
Conditional re-employment wages with selection caveat
Quarterly/weekly dynamics
2021–22 outcomes
Employer/occupation quality
13/26/52-week retention
Confidence intervals and selection bounds

If not feasible

Remove «no subsequent earnings gain» from abstract, introduction and conclusion.
Keep descriptive indices only as secondary context with limited language.

A null descriptive pattern is not evidence of no causal match-quality benefit.

Robustness should target specific failure modes

Threat	High-value check	What would worry us
Seasonality / break mining	Multiple pseudo-dates and 2019/2022 placebos	Similar breaks occur often
Outcome definition	1/2/4-week persistence; fortnightly aggregation	Effect appears only under one cadence
Policy timing	22 Mar, 24 Mar, early Apr, 27 Apr; anticipation windows	Estimate depends on undocumented cut-off
Composition	Balanced/open cohorts; leave-one-group-out	One industry/region/age drives result
JobKeeper	Actual/predicted exposure splits	Exit effect concentrated in supported jobs
Matching	Calipers, replacement, overlap/entropy weights	Poor overlap or sign instability
Parallel trends	Long pre-period + HonestDiD	Small plausible drift overturns result

Summarise a pre-specified battery in a robustness table/specification curve; avoid dozens of selectively narrated regressions.

The main paper needs a small set of decisive empirical exhibits

#	Exhibit	Question answered
1	Institutional timeline + eligibility diagram	Who is exposed to which policy component and when?
2	Cohort and balance table + Love plot	Are groups comparable on pre-policy variables and outcomes?
3	Long event-study gap figure	Do paths align before the event; when does the gap open and close?
4	Main person-week effects table	What is the effect under exact risk sets and conservative inference?
5	Threats / robustness table	How sensitive are results to JobKeeper, dates, trends and transition definitions?
6	Predicted replacement-rate figure	Does response scale with actual policy intensity?
7	Model fit / held-out range / interaction decomposition	Which mechanisms are quantitatively consistent with the revised moments?

Everything else—matching variants, placebo grids, subgroup tables and model derivations—can move to the online appendix.

05

From analysis to a rewritten paper

Narrow the claim, lead with uncertainty and make every design choice auditable.

A defensible contribution has three parts

Empirical design and result

Linked weekly administrative records compare pre-policy-similar Australian and ineligible New Zealand citizens in shared labour markets; estimate transitions with design-specific uncertainty and explicit bundle caveats.

Mechanism and broader lesson

A partial-equilibrium dynamic model shows how a flat benefit increase and residual work-surplus wedge can interact; emergency benefit effects need not equal normal-times elasticities.

Do not oversell «clean identification», «identical conditions», voluntary separations, isolated supplement effects or welfare.

Recommended paper architecture

Section	Job
1. Introduction	Question, comparison, revised estimates with intervals, model range, scope limits
2. Institutional setting and timeline	Eligibility, bundle, JobKeeper, obligations, announcement/payment dates
3. Data and cohort construction	Pre-policy universe, linkage, weekly states, risk sets, balance and coverage
4. Empirical strategy and inference	Estimand, person-week hazards, time-series uncertainty, HonestDiD
5. Effects on transitions	Job finding and recorded payroll exits; dynamics and magnitudes
6. Mechanisms and robustness	Policy interactions, replacement rates, earnings, placebos
7. Dynamic interpretation model	Model 15 rewritten, partial identification and interaction decomposition
8. Conclusion	One broad lesson, uncertainty and explicit limits

Rename the «RDD» material as controlled interrupted time-series / event-time robustness; time is not locally random at the pandemic cut-off.

Results-writing discipline

Use

- Percentage points first
- Baseline rate and named denominator
- Confidence interval in prose
- «Estimated» / «consistent with»
- Recorded payroll-employment exit
- Policy exposure / JobSeeker expansion
- Fitted versus held-out model moments

Avoid

- Significance stars as evidence
- Non-significance = zero
- Comparing subgroup significance
- «Identical conditions»
- Voluntary separation without exit reason
- Supplement-only causal label
- Health or welfare labels not identified

A careful vocabulary will strengthen rather than weaken the paper: it tells referees the authors understand the design's exact reach.

Execution sequence and submission gates

Gate	Work	Pass condition	If it fails
G1 Cohort/risk sets	Rebuild without future receipt or non-employment	Effects remain economically meaningful	Reframe as descriptive or redesign
G2 Inference	One-policy serial-correlation package	Intervals/placebos still support central claim	Do not present conventional 5% causality
G3 Policy isolation	JobKeeper + policy bundle tests	Stable estimates under credible exposure controls	Label bundle; narrow exit claim
G4 Model discipline	Recalibrate revised moments; admissible set	Qualitative mechanism survives near-fitting set	Use model only illustratively
G5 Rewrite	New outline, consistent claims and numbers	Internal referee can state contribution in two sentences	Delay submission

Do not begin final prose polish until G1–G3 are passed; otherwise every headline number and structural target may move.

Likely seminar and referee questions

- Why should Australian and New Zealand citizens have parallel counterfactual paths during a border closure and public-health shock?
- How can treatment be future JobSeeker receipt if receipt depends on the labour-market outcome?
- Why does the employment-exit sample condition on eventually becoming non-employed?
- What is the effective number of policy shocks, and why are the standard errors valid?
- How do JobKeeper, eligibility waivers and mutual obligations map into the treatment?
- Does a missing STP payment represent unemployment, zero hours or reporting cadence?
- Why do high-income bins show the largest JFR response under a flat supplement?
- Which model moments are truly held out and what survives across near-fitting specifications?

A strong paper answers these questions in the design and exhibit choices—before the referee report.

Take-home messages

Keep

The citizenship-based institutional comparison, linked weekly administrative data and focus on employment transitions.

Rebuild

Treatment assignment, cohort/risk sets, person-week estimation, long pre-trends, one-shock inference, JobKeeper exposure and actual replacement rates.

Reframe

Emergency income support and employment transitions under weak demand—not a clean supplement-only welfare verdict.

If the redesigned results remain stable, this can become a compelling labour/public economics paper. The next marginal hour belongs in cohort and inference code, not extra robustness prose.